

A Curated Web Dataset of Mbala Municipal Council Publications, CDF Projects and Local Development Activities

CSC 4792 Data Mining and Warehousing — Mini Project
Project Team 17 — Mbala Municipal Council

Data Description

This dataset was prepared for the CSC 4792 Data Mining and Warehousing mini-project. The assigned council is Mbala Municipal Council (Project Team 17). The project curates page-level information from the supplied extract of the council's official website and structures explicit development-related records for analysis and reuse.

Specifications

Subject area	Data Mining and Warehousing
Specific subject area	Municipal web data, local development and CDF/LGEF activities
Type of data	Web-page text records and structured development records
Geographical coverage	Mbala District, Northern Province, Zambia
Source	Mbala Municipal Council official website and supplied scraped extract
Page-level records	42 records in the supplied source extract
File format	CSV using pipe () delimiter

Background and Motivation

Local government institutions publish information about public services, projects, budgets, notices and community development activities through digital channels. This project curates such information into a reusable dataset for data mining and exploratory analysis. The project specification identifies CDF allocations and projects, approved budgets and revenue information, Integrated Development Plans, ward and administrative information, and council reports or resolutions as important categories of municipal information.

Data Sources and Collection

The source material consisted of scraped page-level records from the Mbala Municipal Council website. The source CSV contained URLs, page/post identifiers, titles, dates, topic labels and extracted text. The original source URL is retained in the dataset to support provenance and future enrichment. A structured CDF/LGEF dataset was also curated from explicit project, asset, empowerment and bursary information present in the supplied records. Only information explicitly supported by the source was retained; missing values were not estimated.

Data Processing and Cleaning

The processing workflow was implemented in Python using pandas and documented in a Jupyter Notebook. It included loading the pipe-delimited source file, inspecting its structure, standardising identifiers, converting dates, normalising whitespace in text fields, recalculating text length, profiling topic labels, and performing validation checks. Structured development records were additionally checked for valid numeric amounts, progress percentages and beneficiary/unit counts.

Data Structure

The cleaned page-level dataset contains the following fields:

Field	Description
url	Original Mbala Municipal Council page URL.
page_id	Council page identifier, where available.
post_id	Council post identifier, where available.

Field	Description
title	Title of the page or post.
date	Publication date, standardised where available.
topics	Topic labels associated with the record.
text_content	Extracted textual content from the page or post.
text_length_chars	Number of characters in the cleaned text content.

The structured CDF/LGEF dataset uses one record for each explicit project, asset, empowerment or bursary item and includes fields for record identifier, category, project/programme description, ward, CDF year/period, amount in Kwacha, reported progress, beneficiaries or units, sector, and source URL.

Quality Control and Validation

Quality control included checking record counts, duplicate URLs, duplicate structured record identifiers, missing values, and numeric fields. The final CSV files use the pipe character (|) as required by the project specification. Source URLs are retained so individual records can be traced to their originating council pages.

Selected Findings

The supplied source contains examples of infrastructure, market, water and sanitation, CDF empowerment, bursary and related local development activities. Explicit project records include a President Avenue CBD facelift reported at K8.7 million and 30% progress, a Mbulu Modern Market Shelter reported at K6.8 million and 65% progress, and a Kawimbe Secondary School waterborne ablution and mechanised water reticulation project reported at K1,058,557.10 and 80% progress.

Potential Reuse Value

The dataset can support exploratory data analysis, municipal information retrieval, CDF project profiling, sector classification, cost and progress analysis, topic profiling, text mining and topic modelling. Retained source URLs also support future enrichment with additional council documents where those underlying documents are accessible.

Limitations

The supplied CSV is a page-level web extract and is not a complete archive of every document linked from the council website. Some pages identify downloadable documents without including their full contents in the supplied extract. Consequently, this dataset does not claim to contain complete council budgets, full Integrated Development Plans, complete Ward Development Committee records, or every CDF beneficiary or project. Missing values were not estimated or fabricated.

Ethical and Provenance Considerations

The dataset is based on publicly available municipal web information supplied for the academic project. Source URLs are retained for provenance. Values are reported only when supported by the source extract, and empty fields indicate that the relevant information was not available in the supplied source.

References

1. CSC 4792 Data Mining and Warehousing Mini Project Practical Assignment, University of Zambia, September 2026.
2. Mbala Municipal Council. Official website: <https://www.mbalacouncil.gov.zm/>
3. Republic of Zambia. Local Government Act No. 2 of 2019; Local Government (Amendment) Acts 2023 and 2026.
4. Phiri, L. (2026). A Multi-Source Dataset for CS1 Failure Prediction [Dataset]. Kaggle.